Document Chatbot (RAG)

Project Overview

This project is an advanced **Retrieval-Augmented Generation (RAG)** system. The system is designed to act as an intelligent assistant that provides accurate answers based strictly on an internal operational manual. It eliminates AI hallucinations, ensures data privacy by running locally, and provides transparent source tracking for every response.

Technical Stack

- **Language:** Python 3.10+
- **LLM Orchestration:** LangGraph
- **Vector Database:** Qdrant (Raw Client integration)
- **LLM Engine:** Ollama (Local execution)
- **Models:**
  - **Generation:** `llama3.2` (3B parameters)
  - **Embeddings:** `nomic-embed-text` (768 dimensions)
- **Document Processing:** PyPDF RecursiveCharacterTextSplitter

Chosen Architecture

The system utilises a modular **Graph-based Architecture** instead of a linear chain.

- **Retrieve Node:** Performs a semantic search using the Qdrant raw client to fetch the most relevant document chunks based on cosine similarity.
- **Generate Node:** Processes the retrieved context along with conversational history to formulate a point-wise response.
- **State Management:** A centralised **`TypedDict`** state ensures data consistency across the retrieval and generation phases.

Key Features & Implementation Details

Strict Hallucination Control

The system is engineered with a "Knowledge-First" approach. Through rigorous system prompting and temperature control (Temperature=0), the model is restricted from inventing facts. If the information is missing from the vector store, the system triggers a mandatory fallback response: *"This information is not present in the provided document."*

Conversational Memory (Context Persistence)

To support follow-up questions (e.g., "What is the citation for the last answer?"), The system maintains a manual sliding window memory of the **last 4 conversation pairs (8 messages)**. This provides enough context for the LLM to understand pronouns and references without overloading the context window.

Source Citation & Similarity Scores

For enhanced transparency, every point in the AI's response is appended with:

- **Similarity Score:** The mathematical confidence level from the Qdrant search.
- **Page Citation:** The exact page number from the original PDF, extracted via metadata.

Justification of Technical Decisions

- **Why LangGraph?** It provides a robust framework for building stateful, multi-actor applications with loops and conditional logic, offering better control than standard LangChain.
- **Why Qdrant Client?** It is a lightweight and reliable system. I can use it in my local machine and store vector embeddings locally.
- **Why Local LLM (Ollama)?** This ensures 100% data privacy for internal company documents, as no data ever leaves the local environment to hit third-party APIs like OpenAI.

How to Run

1. **Prerequisites:**

   i) Install Ollama

   ii) Install Qdrant (Docker recommended).

2. **Setup:** run the following commands in the terminal

   i) pip install -r requirements.txt

   ii) ollama pull llama3.2

   iii) ollama pull nomic-embed-text

3. **Execution:**

   i) Provide a valid pdf path in the database_create.py file and run it.

   ii) Run **uvicorn main:app - -reload**

   iii) Run **streamlit run app.py**

do they maintain inclusive environment?

Yes, they do. According to the provided context, they adhere to the "Inclusive Language Guide v4.0" which includes:

- Pronoun Respect: They respect individual identity and encourage employees to list their pronouns in Slack and Email signatures.
- Neutral Terminology: They avoid gendered idioms and use "Primary/Secondary" instead of "he/she" or other gendered terms.

Additionally, they also have a "No-Meeting Days" policy to allow for deep-work blocks and a "Hot-Desking Policy" to maintain a professional workspace.

leave policy

According to the provided context, the leave policies are as follows:

- Annual Leave: 20 days. Unused leave expires on March 31st of the following year.
- Sick Leave: 12 days. Mental Health days are categorized under Sick Leave; no explanation is required for a 1-day absence.
- Maternity/Paternity: Primary caregivers receive 100% pay for 16 weeks. Secondary caregivers receive 100% pay for 6 weeks.

what is the page no for your last message

Score: 0.509, Page: 2

what is their leave policy?

Their leave policy is as follows:

● Annual Leave: 20 days. Unused leave expires on March 31st of the following year. ● Sick Leave: 12 days. Mental Health days are categorized under Sick Leave; no explanation is required for a 1-day absence. ● Maternity/Paternity: Primary caregivers receive 100% pay for 16 weeks. Secondary caregivers receive 100% pay for 6 weeks.

do they provide extra money for over time?

Confidence score: 0.8 and Page: 4

According to the provided context, overtime is compensated at 1.5x the hourly rate or as "Time-Off in Lieu" (TOIL). There is no mention of an additional bonus for overtime.